

BALANCE ASSESSMENT OF DISC GOLF PLAYERS BY A SMARTPHONE

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ABSTRACT

Balance Assessment of Disc Golf Players by a Smartphone

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Disc golf is an underrepresented sport in biomechanics research, even though it could have effects on gait and balance. Forward and side lunges performed while throwing discs can improve static and dynamic balance. The purpose of this study was to explore the effects of playing disc golf on various static and dynamic balance metrics. Static and dynamic balance experiments were done on 20 disc golf players and 20 control participants to record their RMS of acceleration during the double leg, tandem, and single leg static poses, as well as their gait speed, cadence, stride length, right step length, left step length, and step width during gait. Static data were collected using the acceleration of a handheld smartphone using the MATLAB app, and dynamic data were collected with the smartphone in the participant's pocket using the OneStep app. The results of the disc golf players were compared against those of the control group using t-tests and regression analyses. It was found that (1) the disc golf players had improved gait speed, stride length, right step length, and left step length, and (2) after accounting for age, it was found that older age was correlated with worse tandem balance. The findings showed that playing disc golf can improve dynamic balance, but not necessarily static balance.

Keywords: Disc golf, biomechanics, balance, gait, smartphones

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TABLE OF CONTENTS

	Page
LIST OF TABLES	viii
LIST OF FIGURES	ix
CHAPTER	
1. INTRODUCTION	1
2. METHODS	5
2.1 Participant Demographics	5
2.1.1 Control Experiments	5
2.1.2 Player Experiments	7
2.2 Experiments	8
2.2.1 Static Test Protocol	8
2.2.2 Dynamic Test Protocol	12
2.3 Post-Processing	14
2.3.1 Static Data Processing	14
2.3.2 Dynamic Data Processing	15
2.4 Statistics	15
3. RESULTS	17
3.1 Disc Golf Players vs Control	17
3.2 Regression Analysis with Age	18
4. DISCUSSION	20
4.1 Summary of Key Findings	20
4.2 Limitations	23
4.3 Future Research	24
4.4 Conclusion	25
REFERENCES	26

APPENDICES

A: ORIGINAL DATA	30
B: MATLAB SOFTWARE	31
C: MBL PROTOCOL FOR BALANCE EXPERIMENTS	32
D: PARTICIPANT DATA, STATIC CONTROL GROUP	39
E: PARTICIPANT DATA, STATIC DISC GOLF GROUP	40
F: PARTICIPANT DATA, DYNAMIC CONTROL GROUP	41
G: PARTICIPANT DATA, DYNAMIC DISC GOLF GROUP	42
H: BAR GRAPHS OF RESULTS	43

LIST OF TABLES

Table	Page
1. Control group characteristics.	5
2. Player group characteristics	7
3. Dynamic gait variables	15
4. Means and standard deviations of data with t-test results. Statistically significant differences marked with an “*”.	17
5. P-values for regression models including age.	18
6. Partial p-values of each significant parameter.	19
7. General equations of each significant parameter.	19
D-1. Static data of the control group participants.	39
E-1. Static data of the disc golf group participants.	40
F-1. Dynamic data of the control group participants.	41
G-1. Dynamic data of the disc golf group participants.	42

LIST OF FIGURES

Figure	Page
1. Hallway that most dynamic tests of the control group were done in.....	6
2. Walkway at Sinsheimer Park	8
3. MATLAB Application Sensor UI.	9
4. Double Leg Stance.....	10
5. Tandem Stance	10
6. Single Leg Stance	11
7. OneStep App UI.....	13
C-1. Hallway outside MBL for use in dynamic experiments.	33
C-2: Sensors tab in the MATLAB app.....	34
C-3. Double leg, tandem, and single leg stances.	35
H-1. RMS acceleration in the double leg stance of both groups (p=0.8676)	43
H-2. RMS acceleration in the tandem stance of both groups (p=0.9968)	43
H-3. RMS acceleration in the single leg stance of both groups (p=0.9891).....	44
H-4. Mean gait speed of both groups (p=0.0014) * = significant.....	44
difference (p<0.05)	44
H-5. Mean cadence of both groups (p=0.2145).	45
H-6. Mean stride length of both groups (p=0.0037) * = significant.....	45
difference (p<0.05)	45
H-7. Mean right step length of both groups (p=0.0028). * = significant.....	46
difference (p<0.05)	46
H-8. Mean left step length of both groups (p=0.0094). * = significant.....	46
difference (p<0.05)	46
H-9. Mean step width of both groups (p=0.9413).	47

1. INTRODUCTION

Disc golf is a sport played by hundreds of thousands of players worldwide [1], yet the biomechanics of the sport have rarely been studied. The sport includes players with a wide age range, with 34% of registered Professional Disc Golf Association (PDGA) players in 2023 being above 40 years of age [2]. The sport involves repetitively throwing discs, which has the thrower perform an in-line or side lunge. Studies have shown that these types of lunges are a good exercise for improving overall balance (i.e., postural stability). Performing lunges strengthens the muscles in the lower extremities, which increase postural stability during balance poses. In-line lunges were shown to improve mediolateral balance more so than regular lunges [3], and an 8-week lower limb exercise program including side lunges was shown to increase both static and dynamic balance [4]. A program including side lunges was also linked to improved balance among professional fencers [5].

Disc golf is played similarly to standard golf; participants attempt to throw a Frisbee-like disc into a distant basket in as few throws as possible. Players generally begin by throwing a disc from the tee area, and then they walk to the disc and attempt to throw it to the basket. Many courses have a par of 3 for each hole, which means that on average it should take 3 throws to get the disc in the basket. During one 18-hole disc golf game, players would throw about 54 discs, assuming an average par of 3 per hole. While it is likely that not all these throws involve a lunge, at least the first throw of each hole would include one as players tend to maximize power on the first throw, resulting in at least 18 lunges. The workout program referenced in [4] had the participants do about 50

repetitions of front and side lunges per week for 8 weeks, so it can be estimated that balance improvements from playing disc golf would start to be visible in a similar amount of time for players that throw multiple discs from the tee or those that play multiple rounds a week. Therefore, it is possible that the number of lunges that disc golf players perform whenever they play influences their overall balance.

Finding methods for the older population to improve their balance is very important. Among adults over 65, unintentional falls are one of the leading causes of injuries and death. In 2019, a study found that 27.6% of adults reported falling, with over 38,000 people dying as a direct result of falls in 2021 [6]. Physical activity is a good way to reduce the risk of falls and stay fit, but it can be difficult for older people to participate in most sports due to joint pain and reduced cardiovascular fitness. However, due to disc golf's low intensity, the sport can be an attractive and accessible option for older individuals with balance disorders who seek to improve their balance.

Apps such as OneStep can be used by doctors, physical therapists, and individuals who wish to self-monitor as an option to quantify dynamic balance by tracking related gait metrics. Since the app can be downloaded on smartphones, it is a much more accessible method to track gait metrics than going into a motion analysis lab. The app tracks several gait metrics that have been linked to improved dynamic balance.

Gait speed and cadence have been shown to be indicators of balance, as a study found that people in the lower percentiles of gait speed had a higher prevalence of balance disorders [7] and faster cadence has been shown to be associated with fewer falls per step [8]. Similarly, shorter stride length was found to be associated with more balance-related adverse clinical events [9]. Studies have found that step length shortens

with age in order to counteract reduced balance ability, which could indicate that individuals with longer step lengths have greater natural balancing ability [10]. It is important that both right and left step lengths are measured, as a significant difference between them can be an indicator of asymmetrical gait [11]. A study in healthy young adults found that a higher step width was associated with decreased balance control [12]. OneStep has been validated against a motion analysis system to measure these variables accurately [13].

The current gold standard for measuring gait and balance is the distance traveled of the center of pressure during a laboratory-based balance test but measuring this comes with many limitations. Balance tests require significant setup and need the participant to go to a motion analysis lab, which can make it difficult for patients to consistently measure their balance. Personnel must be present for experiments and be trained in equipment operation and data postprocessing. Furthermore, doing tests in a lab may not replicate real-world conditions, so the data acquired may be inaccurate [14]. This is especially important in dynamic balance tests, as walking conditions are not always as flat as they are in motion analysis labs.

Measuring balance using smartphones can be a convenient way for people to self-monitor their balance and track their progress as they exercise. Smartphones are portable and have a more familiar user interface, which makes it more likely that patients would properly run the tests. Studies have shown that inertial measurement units (IMUs) in smartphones can be a reliable way to measure gait metrics [15]. These IMUs can measure the acceleration of the smartphone, and the root mean square (RMS) of this acceleration can be used as an indicator for static balance [16] [17]. During a balance test, RMS shows

the average magnitude of the participant's acceleration during the pose, which quantifies how much they are swaying to keep themselves balanced. The acceleration is measured in three standard static balance poses: the double leg, tandem, and single leg stances. These poses can be used to track the movement of the center of pressure of a participant, which is the gold standard for assessing static balance [18].

The objective of this study was to determine how static and dynamic balance metrics differ between disc golf players and non-disc golf players. Measures of static balance were the RMS acceleration measurements of three balance poses: double leg stance, tandem stance (dominant foot behind the other, heel-to-toe position), and single leg stance [19]. Measures of dynamic balance included gait speed, cadence, stride length, right step length, left step length, and step width. While stride length, step length, and gait speed are not direct measurements of balance, previous studies have shown that they are related to balance and stability while walking [20], [21]. The two hypotheses of this study were that disc golf players on average would have improved 1) static (RMS acceleration in double leg, tandem, and single leg poses) and 2) dynamic (gait speed, cadence, stride length, right step length, left step length, step width) balance metrics when compared against a control group.

2. METHODS

Protocols were approved by the Cal Poly institutional review board and were designed to minimize risks to participants. The official Mobile Biomechanics Lab (MBL) protocol for smartphone balance experiments is included in Appendix C.

2.1 Participant Demographics

2.1.1 Control Experiments

Control experiments consisted of 20 adults sampled from California Polytechnic State University, San Luis Obispo. The inclusion criteria for the tests were the following: college-aged students aged 18-80 years and English speaking. The exclusion criteria were the following: history of cardiovascular, respiratory, or metabolic disease, history of orthopedic pathology, medical conditions that prohibit physical activity, or pre-existing joint injuries that impact walking ability. The demographics of the participants are recorded in Appendix D and summarized in Table 1.

Table 1. Control group characteristics.

Variable	Value
Age (years)	21.2±1.09
Sex (female/male)	9/11

Control experiments were done mostly in the Cal Poly Mobile Biomechanics Lab (MBL, Building 52, Room D17), while some were performed on a street (San Luis Obispo, CA). Static tests were done inside the MBL, and dynamic tests were performed in the building hallway outside the lab (Figure 1). Static tests were done before dynamic tests.



Figure 1. Hallway that most dynamic tests of the control group were done in.

Participants were instructed to come to experiments wearing pants that had front pockets.

Participants were briefed on the purpose and protocol of the experiments and signed informed consent, participant information, and photographic release forms. Static and dynamic tests were performed, and the data were shown to participants if requested.

2.1.2 Player Experiments

The disc golf player group consisted of 20 adults sampled from local disc golf courses. The inclusion criteria were the following: English speaking adults aged 18-80, at least one year of disc golf experience, and usually played at least once a week. The exclusion criteria were the following: history of cardiovascular, respiratory, or metabolic disease; history of orthopedic pathology; medical conditions that prohibit physical activity; or pre-existing joint injuries that impact walking ability. The age and sex of the disc golf players are recorded in Appendix E and summarized in Table 2.

Table 2. Player group characteristics

Variable	Value
Age (years)	37.4±12.9
Sex (female/male)	0/20

Experiments were performed at the courses in Sinsheimer Park (San Luis Obispo, CA) and Waller Park (Santa Maria, CA) on surfaces as flat as possible (Figure 2). Participants were briefed on the purpose and protocol of the experiments and signed informed consent, participant information, and photographic release forms. Static and dynamic tests were performed, and the data were shown to participants if requested. Static tests were done before dynamic tests.



Figure 2. Walkway at Sinsheimer Park

2.2 Experiments

2.2.1 Static Test Protocol

The MATLAB (MathWorks, Natick, MA) smartphone application was downloaded onto the MBL iPhone 11 (Apple, Cupertino, CA) prior to data collection. The MATLAB app uses the iPhone's built-in accelerometer to record the acceleration, velocity, and orientation of the phone. The "Sensors" section of the MATLAB app was used to record the data with the sampling frequency set to 100 Hz (Figure 3).

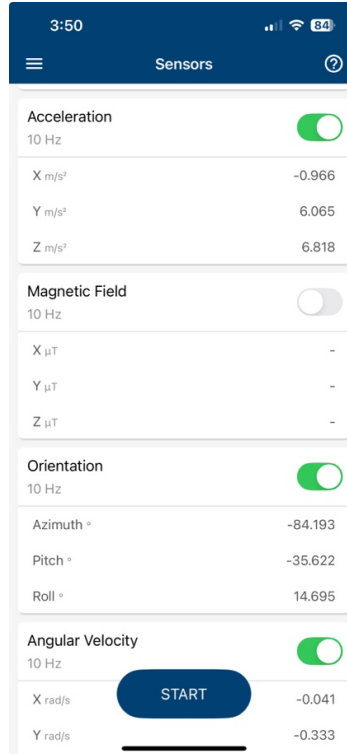


Figure 3. MATLAB Application Sensor UI.

Participants were briefed about the three balance poses that they would be performing in the static tests in the following order: the double leg, tandem, and single leg stances. The double leg stance had the participant stand with both feet shoulder width apart (Figure 4), the tandem stance had the participant place their dominant foot behind the other in a straight line (Figure 5), and the single leg stance had the participant stand on their non-dominant leg (Figure 6). After starting the MATLAB sensors, participants were instructed to hold the phone in their left hand and cross their arms over their chest before holding the pose.



Figure 4. Double Leg Stance



Figure 5. Tandem Stance



Figure 6. Single Leg Stance

Each pose was held for 30 seconds, which was timed by the researcher using a stopwatch. At the beginning and end of each recording, the participant was instructed to do a small jump. The purpose of the jumps was to form a spike in the acceleration data, making it easier to identify the beginning and end of the pose. The data were inspected after each test to confirm all variables were properly recorded before moving on to the next pose. One recording was made for each pose.

2.2.2 Dynamic Test Protocol

The OneStep (OneStep, New York, NY) smartphone application was downloaded onto the MBL iPhones prior to data collection. It is not clear in the OneStep app whether the data collected are filtered before the results are displayed.

The 1-minute walking test (Figure 7) on OneStep, which has the participant walk at a self-selected gait speed in a straight line for one minute, was used to record gait speed, cadence, stride length, right step length, left step length, and step width. The OneStep app has previously been validated against a motion analysis system to record these metrics [13]. Variables from the OneStep report were recorded in a Microsoft Excel (Microsoft, Redmond, WA, USA) spreadsheet on a laptop.

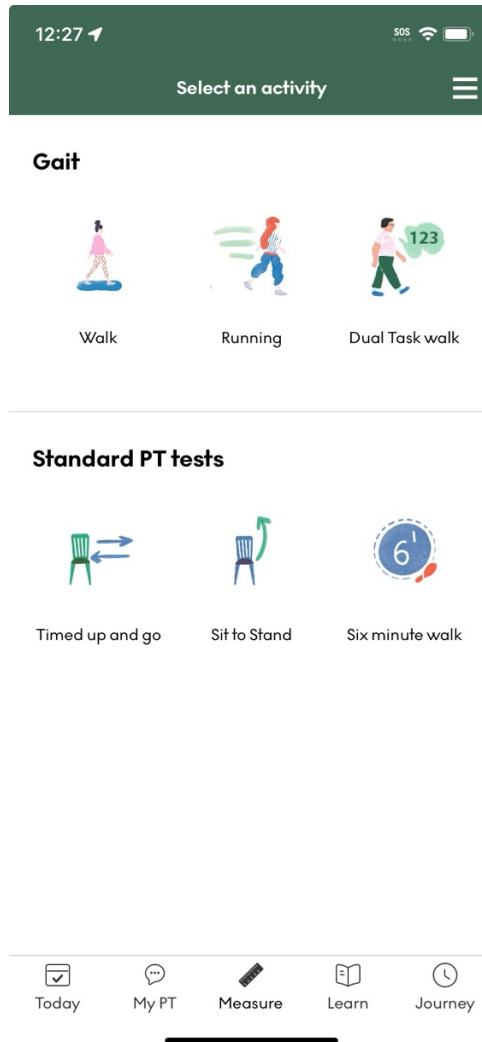


Figure 7. OneStep App UI.

At the start of each trial, the participants pressed the “start countdown” button in the application and placed the phone into their right front pants pocket. The phone was placed vertically such that the long axis of the phone was aligned with the long axis of the leg. The position of the phone was verified before the test began. During the 1-minute walking test, the participants were instructed to perform the serial sevens test, in which they repeatedly subtracted 7 from 100. The purpose of this was to prevent the Hawthorne effect, which is when the gait of participants is influenced by the presence of an observer

[22]. By giving participants a subtraction task, they were less focused on the experiment and had a more normal gait. The test ended after one minute when the app audibly announced the test was over. If the end of the hallway was reached before 1 minute was up, participants were instructed to make wide turns while doing their best to keep the same gait speed. The test was repeated two more times for each participant (for a total of three trials per participant), with the serial sevens test starting at 101 and 102 for each subsequent test to avoid repetition.

2.3 Post-Processing

2.3.1 Static Data Processing

A MATLAB script developed by MBL students was used to postprocess the static data [23] [24] [25]. The array that the MATLAB app recorded during the static tests was initially converted into a .csv file through the script. Since MATLAB includes the acceleration from gravity in its recording and the analyses needed the raw acceleration, the acceleration vector was subtracted from the raw data in the script. More information about the gravity subtraction and the equations used are included in Appendix C. The .csv file was then put into a postprocessing algorithm, which displayed a plot of the data. The script allowed for heel strike and toe-off to be identified in the capture and then trimmed the capture from three seconds after heel strike to three seconds before toe-off. This was to ensure that any acceleration data from during the two jumps were not included in the analysis. The recording was then filtered using a 4th order Butterworth filter with a cutoff frequency of 3.5 Hz, and the root mean square (RMS) of the acceleration magnitude was calculated and recorded.

2.3.2 Dynamic Data Processing

Postprocessing of the data collected during the dynamic trials was automatically processed in the OneStep app. The report of each OneStep trial was generated to obtain all the variables of interest, which are summarized in Table 3. The average of each variable across the three trials for each participant was used for analysis.

Table 3. Dynamic gait variables

Variable	Units	Definition
Gait Velocity	ft/s	The velocity of the participant's gait
Cadence	steps/min	The number of steps taken per minute
Stride Length	in	The distance between consecutive heel strikes of one foot during the gait cycle
Right Step Length	in	The distance between the heel strike of the right foot from the left during the gait cycle
Left Step Length	in	The distance between the heel strike of the left foot from the right during the gait cycle
Step Width	in	The distance between heels during the gait cycle

2.4 Statistics

T-tests were performed on each measured metric to see if there was statistical significance between the disc golf players and the control group. Also, linear regressions were performed on the data to include age as a variable to see if the age of the

participants had a significant effect on their balance. Significance was defined by $p < 0.05$. All statistical analyses were performed in JMP (JMP Statistical Discovery, Cary, NC).

3. RESULTS

3.1 Disc Golf Players vs Control

The results of the t-tests showed that there were statistically significant differences in gait speed ($p=0.0014$), stride length ($p=0.0037$), right step length ($p=0.0028$), and left step length ($p=0.0094$) (Table 4). On average, disc golf players were observed to have a faster gait speed, a longer stride length, and longer left and right step lengths than the control group. There were no significant differences in step width, cadence, or RMS acceleration in any static pose. Bar graphs of the results are included in Appendix H.

Table 4. Means and standard deviations of data with t-test results. Statistically significant differences marked with an “*”.

Metric	Control Mean $\pm$ Standard Deviation	Disc Golf Mean $\pm$ Standard Deviation	p-value
Double Leg RMS Acceleration (m/s^2)	0.105 ± 0.059	0.127 ± 0.059	0.8676
Tandem RMS Acceleration (m/s^2)	0.126 ± 0.052	0.177 ± 0.053	0.9968
Single Leg RMS Acceleration (m/s^2)	0.151 ± 0.068	0.308 ± 0.253	0.9891
Gait Speed (ft/s)	4.058 ± 0.452	4.429 ± 0.222	0.0014*
Cadence (steps/min)	105.133 ± 6.664	106.617 ± 4.933	0.2145
Stride Length (in)	55.55 ± 4.733	59.4 ± 3.777	0.0037*
Right Step Length (in)	27.667 ± 2.466	9.8 ± 2.107	0.0028*
Left Step Length (in)	27.817 ± 2.496	29.583 ± 2.023	0.0094*
Step Width (in)	5.083 ± 0.996	5.567 ± 0.905	0.9413

*Indicates Statistical Significance

3.2 Regression Analysis with Age

To test for any possible effects of age on the measurements, a regression analysis using JMP was performed. An analysis was performed on each variable using group, age, and the interaction between them as model effects. The p-value for the model utility of each fit is included in Table 5.

Table 5. P-values for regression models including age.

Variable	Fit p-value
DL RMS Acceleration	0.7145
TL RMS Acceleration	0.0002*
SL RMS Acceleration	0.0992
Gait Speed	0.0073*
Stride Length	0.0109*
Cadence	0.4316
Step Width	0.2392
Right Step Length	0.0225*
Left Step Length	0.0118*

*Indicates statistical significance

Of the significant regression models, the partial p-values for each parameter are included in Table 6, and the general equation for each parameter is included in Table 7. The control group is indicated by a 1 in the [Group] variable, and the disc golf group is indicated by a -1.

Table 6. Partial p-values of each significant parameter.

Variable	Group p-value	Age p-value	Interaction p-value
TL RMS Acceleration	0.0029*	0.0135*	0.0041*
Gait Speed	0.0243*	0.0857	0.1051
Stride Length	0.0465*	0.1284	0.2008
Right Step Length	0.0757	0.1966	0.2419
Left Step Length	0.0452*	0.1113	0.2007

*Indicates statistical significance

Table 7. General equations of each significant parameter.

Variable	General Equation
TL RMS Acceleration	$\text{RMS Acceleration} = 0.395 - 0.127[\text{Group}] - 0.012(\text{Age}) - 0.015[\text{Group}] * (\text{Age} - 29.33)$
Gait Speed	$\text{Gait Speed} = 5.671 - 0.711[\text{Group}] - 0.066(\text{Age}) - 0.062[\text{Group}] * (\text{Age} - 29.10)$
Stride Length	$\text{Stride Length} = 72.833 - 7.461[\text{Group}] - 0.684(\text{Age}) - 0.572[\text{Group}] * (\text{Age} - 29.25)$
Right Step Length	$\text{Right Step Length} = 35.714 - 3.645[\text{Group}] - 0.318(\text{Age}) - 0.288[\text{Group}] * (\text{Age} - 29.25)$
Left Step Length	$\text{Left Step Length} = 37.19 - 3.903[\text{Group}] - 0.3737(\text{Age}) - 0.298[\text{Group}] * (\text{Age} - 29.25)$

4. DISCUSSION

4.1 Summary of Key Findings

To our knowledge, this was the first study to investigate the effects of playing disc golf on gait and balance. The goal of the study was to see whether disc golf players would have improved static and dynamic balance metrics. Static tests consisted of participants holding different balance poses to measure their acceleration, and dynamic tests consisted of participants walking in a straight line while a smartphone collected data. The control group represented the college-aged subpopulation well and was almost evenly split between males and females. However, the oldest participant in the control group was only 22, which could be unreliable when comparing them to a group with a wider age range.

During the static tests, the disc golf players were hypothesized to have lower RMS accelerations during the double leg, tandem, and single leg stances. The results did not support this hypothesis. There was no statistical difference between the two groups for the double leg pose ($p = 0.265$). However, the tandem and single leg stances showed that the disc golf players had a statistically significant increase in the RMS acceleration ($p = 0.006$ and $p = 0.022$, respectively). These results were validated against a similar study that used a smartphone to record acceleration during the same poses, which found similar acceleration values [26]. A possible explanation for this is that the age range for the disc golf players (21-66) was much larger than the control group (18-22). While performing experiments, it was observed that older players appeared to have more difficulty holding the tandem and single leg poses but were more comfortable doing the dynamic tests.

In order to see the results of age, a regression analysis including age as a variable was done. The regression analysis provided some interesting results. In the static tests, the tandem stance ($p = 0.0002$) showed a significant difference after accounting for age. There was a significant difference in group ($p = 0.0029$), age ($p = 0.0135$), and interaction (0.0041). Effect of age was visibly noticeable during data collection, as older participants struggled much more with holding this pose. This is corroborated with numerous studies that show that static balance worsens with age [27] [28]. Interestingly, by looking at the general equation, the significance of the group showed that the control group had better tandem balance than the disc golf players. It is not clear why this could be the case, and a larger sample size could help further confirm this relationship. Overall, it appears that playing disc golf does not improve static balance metrics.

During the dynamic tests, the disc golf players were hypothesized to have a higher gait speed, cadence, stride length, and right/left step length, and a lower step width. These variables were chosen because they have been validated against a motion analysis system [13]. The results partially supported this hypothesis. There was a statistically significant increase in the disc golf group for gait speed ($p = 0.00014$), stride length ($p = 0.0037$), right step length ($p = 0.0028$) and left step length ($p = 0.0094$). However, there was no statistically significant change in step width ($p = 0.9413$) or cadence (0.2145). The results of the dynamic testing are similar to what other studies have found, a study done on over 500 adults on many of the same variables including gait speed, step width, and step length had similar values for those metrics [29]. Overall, it appears that disc golf players do have some improved dynamic balance metrics so it could be a good physical activity for the general population to improve their dynamic balance.

In the regression analysis, the significance of gait speed, stride length, and left step length remained unchanged between groups, and there was no significance with age. However, despite having significant model utility, right step length was no longer significant between groups. A possible explanation for this is that the variables were correlated with one another in some way, so age and group did not have strong individual effects. A more likely explanation is that the sample size was too low which resulted in low statistical power, because this was not an issue with the left step length.

A similar study was done on ultimate frisbee players that also linked players doing frequent lunges to improved balance [30]. This study used a motion analysis system to track variables such as double support time, single support time, and stance phase time using markers and force plates. These variables, while different than the ones used in this study, can also be used to measure balance. The study found that the female ultimate players studied had improved balance when compared to the control group, with a significant decrease in double support time and stance phase time, and an increase in single support time. These results partially validate the improved dynamic balance metrics among disc golf players, as some improvements were seen in both disc-throwing groups.

A possible explanation for the improvements in balance being less significant than hypothesized is that disc golf players aren't performing full lunges with proper form, which would reduce their improvements on overall balance. Since the sport is relatively slowly paced and frequently played alone, it is possible that there is lesser focus on throwing mechanics, which could result in weaker lunges. Furthermore, older players may not have the strength or stability to fully lunge, reducing the effect of any

improvements to balance. Another reason is that playing disc golf does not directly challenge players' balance like other sports would; the only improvements would be from doing lunges. It is possible that the lunges did more to improve dynamic balance, but not static balance. Studies have shown that significant improvements in balance can come from doing proprioceptive exercises, and since disc golf doesn't necessarily involve those exercises, there may be less improvements to balance [31].

Despite the worse static metrics, the disc golf players still had improved dynamic balance metrics. A possible reason for this improvement is that many disc golf courses are quite large and over a variety of elevations, so many disc golf players would have more experience with uneven terrain. This could mean that the disc golf players would have better dynamic balance, though it would be unrelated to playing disc golf and could be achieved with something like hiking. This is supported by various studies, one of which found that a 12-week trekking exercise improved balance in obese women [32], and another which found that children that played outdoors had improved static balance during a one-legged balance test [33]. Another study found that over time, adults exhibited increased step length and reduced step width as they got more familiar with walking on uneven terrain, which could imply that disc golf players who walk on uneven terrain could show similar improvements [34].

4.2 Limitations

A limitation of this study was the discrepancy between the age ranges of the groups. The control group was comprised of college-aged participants aged 18 to 22, while the disc golf group ranged from 21 to 66. As a result, it is very likely that the relationships found could be a result of aging and not playing disc golf.

Another limitation was that there were no female disc golf players tested in the study. The control group was 45% female and comparing that group to the all-male disc golf group could result in issues in the data. Studies have established that there are differences in balance across sex, such as one that found differences in whole body coordination [35] and another that found differences in dynamic balance among children [36]. Ideally, the groups should have roughly the same sex distribution to ensure accurate data.

A third limitation was the conditions that the disc golf data were collected in. Most of the data were collected during a barbeque immediately after a large regional tournament, so many of the players may have been tired from playing in the tournament. As a result, they could have sore muscles which could lead to worse overall balance.

Another limitation was that a Bonferroni correction was not applied for the statistical analyses. The purpose of the correction is to lower the likelihood of a false positive occurring from doing multiple statistical analyses on the same data set. Since a total of 9 tests were done on this data set (3 static balance poses, 6 gait metrics), a Bonferroni correction of 9 should be used, lowering the desired p-value for significance to 0.0056. This would make it more difficult to achieve significance; the only difference would be that left step length would no longer be significant if this correction was used.

4.3 Future Research

Future research on the topic should use a control group from a wider population. Examples could include non-players that are near the disc golf courses in the parks, which would include many people that are closer to the age range of the disc golf players.

Future studies could also include players from other locations, since it's possible that players in the Central Coast area have different levels of activity than other areas. A larger sample size could also improve the data, as some metrics in this study, especially the static metrics, had very high standard deviations. It is important that future studies also include female disc golf players to address concerns of equity. Additionally, disc golf participants should be tested before they practice to ensure that their measured balance is not affected by exhaustion. Better testing equipment would also yield better data; using a motion capture system to track the center of pressure of participants is a more accurate measure of balance.

4.4 Conclusion

Despite the limitations, the study showed that smartphones and the OneStep app can be used to easily measure static and dynamic balance metrics in the general population and in real-world settings. On average, disc golf players have an improved gait speed, stride length, and right and left step length. The study established that anyone seeking to improve their gait or balance can try disc golf to see improvements. A good sense of balance is important to maintain for one's safety as they get older, so taking measures to improve it is something that everyone should strive for.

REFERENCES

- [1] J. T. Nelson, R. E. Jones, M. Runstrom, and J. Hardy, "Disc Golf, a Growing Sport: Description and Epidemiology of Injuries," *Orthop. J. Sports Med.*, vol. 3, no. 6, p. 2325967115589076, Jun. 2015, doi: 10.1177/2325967115589076.
- [2] "2023 PDGA Year End Demographics Report," PDGA, Sep. 2024. [Online]. Available: https://www.pdga.com/files/2023_pdga_yearend_demographicsreport.pdf
- [3] P. H. Marchetti, M. A. Guiselini, J. J. Da Silva, R. Tucker, D. G. Behm, and L. E. Brown, "Balance and Lower Limb Muscle Activation Between in-Line and Traditional Lunge Exercises," *J. Hum. Kinet.*, vol. 62, no. 1, pp. 15–22, Jun. 2018, doi: 10.1515/hukin-2017-0174.
- [4] J. Lee and J. Kim, "Effects of an 8-week lunge exercise on an unstable support surface on lower-extremity muscle function and balance in middle-aged women," *Phys. Act. Nutr.*, vol. 26, no. 4, pp. 014–021, Dec. 2022, doi: 10.20463/pan.2022.0020.
- [5] T. Kim, S. Kil, J. Chung, J. Moon, and E. Oh, "Effects of specific muscle imbalance improvement training on the balance ability in elite fencers," *J. Phys. Ther. Sci.*, vol. 27, no. 5, pp. 1589–1592, 2015, doi: 10.1589/jpts.27.1589.
- [6] R. Kakara, G. Bergen, E. Burns, and M. Stevens, "Nonfatal and Fatal Falls Among Adults Aged ≥ 65 Years — United States, 2020–2021," *MMWR Morb. Mortal. Wkly. Rep.*, vol. 72, no. 35, pp. 938–943, Sep. 2023, doi: 10.15585/mmwr.mm7235a1.
- [7] F. Oliva-Zapata *et al.*, "Association between Gait Speed and Balance Disorders in Older Adults from 12 High Andean Peruvian Communities, 2013–2019," *Ann. Geriatr. Med. Res.*, vol. 28, no. 3, pp. 291–300, Sep. 2024, doi: 10.4235/agmr.24.0010.
- [8] A. Mihailovic, R. M. De Luna, S. K. West, D. S. Friedman, L. N. Gitlin, and P. Y. Ramulu, "Gait and Balance as Predictors and/or Mediators of Falls in Glaucoma," *Investig. Ophthalmology Vis. Sci.*, vol. 61, no. 3, p. 30, Mar. 2020, doi: 10.1167/iovs.61.3.30.
- [9] I. Bytyçi and M. Y. Henein, "Stride Length Predicts Adverse Clinical Events in Older Adults: A Systematic Review and Meta-Analysis," *J. Clin. Med.*, vol. 10, no. 12, p. 2670, Jun. 2021, doi: 10.3390/jcm10122670.
- [10] D. D. Espy, F. Yang, T. Bhatt, and Y.-C. Pai, "Independent influence of gait speed and step length on stability and fall risk," *Gait Posture*, vol. 32, no. 3, pp. 378–382, Jul. 2010, doi: 10.1016/j.gaitpost.2010.06.013.
- [11] K. K. Patterson, N. K. Nadkarni, S. E. Black, and W. E. McIlroy, "Gait symmetry and velocity differ in their relationship to age," *Gait Posture*, vol. 35, no. 4, pp. 590–594, Apr. 2012, doi: 10.1016/j.gaitpost.2011.11.030.

- [12] L. K. Molina, G. H. Small, and R. R. Neptune, "The influence of step width on balance control and response strategies during perturbed walking in healthy young adults," *J. Biomech.*, vol. 157, p. 111731, Aug. 2023, doi: 10.1016/j.jbiomech.2023.111731.
- [13] C. Thai, "Measuring Gait of Knee Arthroplasty Patients Pre-Op and Post-Op Using a Smartphone," MS Thesis, California Polytechnic State University, San Luis Obispo, 2024.
- [14] J. C. Christensen *et al.*, "The validity and reliability of the OneStep smartphone application under various gait conditions in healthy adults with feasibility in clinical practice," *J. Orthop. Surg.*, vol. 17, no. 1, p. 417, Sep. 2022, doi: 10.1186/s13018-022-03300-4.
- [15] A. Godfrey, S. Del Din, G. Barry, J. C. Mathers, and L. Rochester, "Instrumenting gait with an accelerometer: A system and algorithm examination," *Med. Eng. Phys.*, vol. 37, no. 4, pp. 400–407, Apr. 2015, doi: 10.1016/j.medengphy.2015.02.003.
- [16] M. Sekine *et al.*, "A gait abnormality measure based on root mean square of trunk acceleration," *J. NeuroEngineering Rehabil.*, vol. 10, no. 1, p. 118, 2013, doi: 10.1186/1743-0003-10-118.
- [17] K. Bohlke, M. S. Redfern, A. L. Rosso, and E. Sejdic, "Accelerometry applications and methods to assess standing balance in older adults and mobility-limited patient populations: a narrative review," *Aging Clin. Exp. Res.*, vol. 35, no. 10, pp. 1991–2007, Aug. 2023, doi: 10.1007/s40520-023-02503-x.
- [18] D. C. C. De Abreu *et al.*, "Standing balance test for fall prediction in older adults: a 6-month longitudinal study," *BMC Geriatr.*, vol. 24, no. 1, p. 947, Nov. 2024, doi: 10.1186/s12877-024-05380-9.
- [19] J. M. Williams, C. Dorey, S. Clark, and C. Clark, "The within-day and between-day reliability of using sacral accelerations to quantify balance performance," *Phys. Ther. Sport*, vol. 17, pp. 45–50, Jan. 2016, doi: 10.1016/j.ptsp.2015.04.002.
- [20] Y. J. Xie, E. Y. Liu, E. R. Anson, and Y. Agrawal, "Age-Related Imbalance Is Associated With Slower Walking Speed: An Analysis From the National Health and Nutrition Examination Survey," *J. Geriatr. Phys. Ther.*, vol. 40, no. 4, pp. 183–189, Oct. 2017, doi: 10.1519/JPT.0000000000000093.
- [21] P. M. McAndrew Young and J. B. Dingwell, "Voluntarily changing step length or step width affects dynamic stability of human walking," *Gait Posture*, vol. 35, no. 3, pp. 472–477, Mar. 2012, doi: 10.1016/j.gaitpost.2011.11.010.
- [22] J. Jeon *et al.*, "Influence of the Hawthorne effect on spatiotemporal parameters, kinematics, ground reaction force, and the symmetry of the dominant and nondominant lower limbs during gait," *J. Biomech.*, vol. 152, p. 111555, May 2023, doi: 10.1016/j.jbiomech.2023.111555.

- [23] J. Matties, “Hands-Free Balance Protocol and Development of an iPhone App for Self-Assessment,” MS Thesis, California Polytechnic State University, San Luis Obispo, 2025.
- [24] E. C. Anthony, O. K. Kam, S. M. Klisch, S. J. Hazelwood, and B. Berg-Johansen, “Balance Assessment Using a Handheld Smartphone with Principal Component Analysis for Anatomical Calibration,” *Sensors*, vol. 24, no. 17, p. 5467, Aug. 2024, doi: 10.3390/s24175467.
- [25] B. M. Presley, J. C. Sklar, S. J. Hazelwood, B. Berg-Johansen, and S. M. Klisch, “Balance Assessment Using a Smartwatch Inertial Measurement Unit with Principal Component Analysis for Anatomical Calibration,” *Sensors*, vol. 23, no. 10, p. 4585, May 2023, doi: 10.3390/s23104585.
- [26] S. Pickering, “Using a Smartphone to Measure Static Balance: Verification and Validation,” MS Thesis, California Polytechnic State University, San Luis Obispo, 2023.
- [27] S. Sasagawa, A. Arakawa, A. Furuyama, and Y. Matsumoto, “Age-related changes in static balance in older women aged in their early sixties to their late eighties: different aging patterns in the anterior–posterior and mediolateral directions,” *Front. Aging Neurosci.*, vol. 16, p. 1361244, Apr. 2024, doi: 10.3389/fnagi.2024.1361244.
- [28] T.-T. Lin *et al.*, “Age-Related Influence on Static and Dynamic Balance Abilities: An Inertial Measurement Unit-Based Evaluation,” *Sensors*, vol. 24, no. 21, p. 7078, Nov. 2024, doi: 10.3390/s24217078.
- [29] L. K. Lau *et al.*, “Reference Values of Gait Speed and Gait Spatiotemporal Parameters for a South East Asian Population: The Yishun Study,” *Clin. Interv. Aging*, vol. Volume 15, pp. 1753–1765, Sep. 2020, doi: 10.2147/CIA.S270407.
- [30] E. Nikceovich, “Analysis of Gait Parameters and Knee Angles in Ultimate Frisbee Players: Implications for Balance and Injury,” MS Thesis, California Polytechnic State University, San Luis Obispo, 2023.
- [31] M. G. Karakaya, Hi. Rutbil, E. Akpınar, A. Yildirim, and İ. Ç. Karakaya, “Effect of ankle proprioceptive training on static body balance,” *J. Phys. Ther. Sci.*, vol. 27, no. 10, pp. 3299–3302, 2015, doi: 10.1589/jpts.27.3299.
- [32] S.-J. Kang, “Trekking exercise promotes cardiovascular health and fitness benefits in older obese women,” *J. Exerc. Rehabil.*, vol. 10, no. 4, pp. 225–229, Aug. 2014, doi: 10.12965/jer.140136.
- [33] I. Fjørtoft, “The Natural Environment as a Playground for Children: The Impact of Outdoor Play Activities in Pre-Primary School Children”.
- [34] J. A. Kent, J. H. Sommerfeld, M. Mukherjee, K. Z. Takahashi, and N. Stergiou, “Locomotor patterns change over time when exposed to an uneven surface,” *J. Exp. Biol.*, p. jeb.202093, Jan. 2019, doi: 10.1242/jeb.202093.

- [35] F. Khobkhun and J. Thanakamchokchai, “Biological sex-related differences in whole-body coordination during standing turns in healthy young adults,” *Sci. Rep.*, vol. 13, no. 1, p. 22112, Dec. 2023, doi: 10.1038/s41598-023-49201-2.
- [36] R. Li, M. Liu, J. Zhu, R. Li, H. Zhao, and L. Zhang, “Age and gender differences in static and dynamic balance of Chinese preschool children,” *Front. Physiol.*, vol. 13, p. 1013171, Oct. 2022, doi: 10.3389/fphys.2022.1013171.

APPENDICES

APPENDIX A: ORIGINAL DATA

Original participant gait data were collected for this research. The data include raw acceleration, angular velocity, and orientation for each static tests and processed data in the form of several gait parameters. The data do not contain identifiable participant information. The data are available upon request by contacting Dr. Stephen Klisch at sklisch@calpoly.edu.

APPENDIX B: MATLAB SOFTWARE

MATLAB software was developed for this research. This MATLAB software is available upon request by contacting Dr. Stephen Klisch at sklisch@calpoly.edu.

APPENDIX C: MBL PROTOCOL FOR BALANCE EXPERIMENTS

SMARTPHONE BALANCE EXPERIMENT PROTOCOL

INTRODUCTION

The objective of this protocol is to outline gait and balance experiments using the MATLAB and OneStep apps on a smartphone. This protocol will also highlight set-up.

MBL Researchers. Trained researchers will conduct all experiments with the participants' wellbeing as priority. There must always be at least 2 trained personnel during testing. Should the situation call for it, one staff will perform CPR and one who can acquire nearby EID and call 911.

EQUIPMENT & MATERIALS

Smartphones. MBL iPhones (Apple, Cupertino, CA, USA) will be used to conduct these experiments. They include a built-in accelerometer which is used to record data. Two phones should be present for experiments in the event that one doesn't work properly.

Apps. Two apps are required for data collection. MATLAB (MathWorks, Natick, MA, USA) is used for static experiments, and OneStep (Celloscope, New York, NY, USA) is used for dynamic experiments.

Walkway. Experiments require a walkway where a participant can walk in a straight line for 1 minute. The walkway should be as flat and straight as possible. A good example is the hallway outside the MBL, it usually takes people 1 minute to go from the exit door next to the MBL to the end of the hallway (Fig. C-1).



Figure C-1. Hallway outside MBL for use in dynamic experiments.

METHODS

Preparation. Before an experiment begins, MBL staff should prepare the smartphones for the experiments 30 minutes beforehand. More information is included in the Setup section.

Consent Forms. There are 3 consent forms that should be filled out before experiments. The consent forms are

1. Informed Consent Form
2. Participant Information Form
3. Photographic Release Form

MATLAB Setup. Before capturing any data, ensure that the MATLAB app is set up correctly. Make sure that you are signed into your MATLAB account, or your data will not be exported. To start setup, tap the three bars in the top left corner and go to “Sensors” (Fig. C-2).

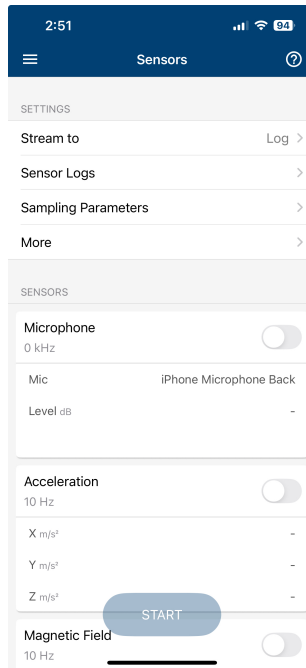


Figure C-2: Sensors tab in the MATLAB app.

Auto Upload. First, go to “More”. Go to the “Upload Folder” tab and make sure that it is going to MATLAB Drive/MobileSensorData. Also go to the “Auto Upload” tab and make sure that it is on. Return to the “Sensors” tab and tab on “Stream to”. Make sure that this is set to Log.

Sampling Parameters. Most of our experiments will sample data at 100 Hz. Go to the “Sampling Parameters” tab and change the motion sensors sample rate to 100 Hz. **This setting resets to 10 Hz every time you close the app, always make sure you change it back to 100 Hz before experiments or you will have bad data.**

Testing Sensors. Scroll down in the Sensors tab and turn on the variables of interest (usually acceleration, angular velocity, and orientation). To make sure that the data upload is working, tap the start button and then stop it. Name the file whatever you want. Go to the “Sensors Logs” tab and wait for the recording to upload, then open MATLAB online on another device. Navigate to the MobileSensorData folder and confirm that your recording is there.

OneStep Setup. You may need to make an account on OneStep before you can use it on the lab iPhones. If it asks if you are a clinician or a patient, select “patient” and create an account with your phone number. Make sure that the sound on the phone is working so that you can hear OneStep’s audio cues. OneStep requires no other setup.

Static Trials. Static tests should be done before dynamic tests. Static trials will use the MATLAB app to measure acceleration during the double leg, tandem, and single leg balance poses. During all poses, the participant will hold the phone in their left hand and cross their arms over their chest, with the arm holding the phone under the other. All poses are pictured in (Fig. C-3.). The poses will be performed in the order below:

Double Leg Stance. The double leg stance will have the participant stand on both feet, with them about shoulder width apart.

Tandem Stance. The tandem stance will have the participant put their feet in a straight line, with the non-dominant leg behind the other. The heel of the front foot should be touching the toe of the front foot.

Single Leg Stance. The single leg stance will have the participant stand only on their non-dominant leg, with their dominant leg raised back at about a 90 degree angle.



Figure C-3. Double leg, tandem, and single leg stances.

Running the trial. Prior to recording, the participant will be briefed about each pose and encouraged to try each of them first. Participants should try each pose 5 times for 30 seconds each. Press the start button on the MATLAB app and give it to the participant. Before starting the pose, the participant should do a small jump. This will cause a spike in the acceleration data, which makes the data easier to postprocess. After the jump, start a timer for 30 seconds. At the end of the timer, ask the participant to jump again, then take the phone and stop the recording. Repeat this for each pose, naming each capture in a way such that you can identify each pose and participant. Go back to your laptop and make sure that the file uploaded and double click it to see how many data points it has. If the app is set to 100 Hz, the 30 second capture should have about 4000-5000 points. If not, the app might be set to 10 Hz.

Dynamic Trials. Go to the walkway and tell the participant that they will walk in a straight line for 1 minute. The participant can practice walking along the walkway to get comfortable with their self-selected speed. The phone will be in the participants right front pocket and be placed vertically. The participant will be doing the serial sevens test while walking, which has them start at 100 and repeatedly subtract 7. This ensures that their gait will be as normal as possible by keeping their mind occupied and not focusing on their gait. In the OneStep app, go to “Measure”, then “Walk”, then “1 minute.” Select “In the pocket” and “Start countdown”. The app will count down from 10, and the participant should start walking at the end of the countdown when the app says to start walking. At the end of one minute, the app will say when to stop. Should the participant reach the end of the hallway, ask the participant to slowly turn to the left while keeping their same gait speed. Repeat this test three times and name each capture in a way such that you can identify the trial and participant. You can view the data by going to the “Journey” tab in the bottom right, tapping on the capture of interest, and scrolling all the way down to “View gait lab report.” So far, we have validated gait speed, cadence, stride length, right step length, left step length, and step width against a motion analysis system. [13].

Postprocessing. To process the static data, the first and last 3 seconds of each capture should be trimmed out to eliminate noise from the jumps during the test. Then, the gravity vector should be subtracted from the recording since MATLAB includes it in its data collection.

Gravity Vector Subtraction. MATLAB records orientation using Euler angles, so the gravity vector must be rotated to subtract it from the acceleration. Begin with a gravity vector of $[0 \ 0 \ 1]$; the units of this vector are in Gs. This is in the global (North East Down) coordinate system. MATLAB records the orientation in the coordinate system relative to the phone, so we want to rotate the gravity vector to the phone’s coordinate system. To do this, we will use the following rotation matrices:

$$\mathbf{R}_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

$$\mathbf{R}_y(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$\mathbf{R}_z(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In these matrices, γ , β , and α correspond to the roll, pitch, and azimuth angles that MATLAB provides, respectively. We can combine these matrices into one transformation using matrix multiplication, to get the following matrix:

$$\mathbf{R} = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_x(\gamma)$$

The order of the multiplication matters, since we are rotating y and x about rotated axes. MATLAB has a built-in function to do this rotation, `eul2rotm`, which uses an input of the orientation data in radians. The gravity vector should be multiplied by the rotation matrix to get the gravity vector in the phone axes. Example code to subtract gravity is included below.

```
% calculate gravity vector in device axes:
gNED = [0 0 1]; % gravity in g in global (NED) axes
gBody = zeros(length(sensorData.Orientation),3); %
initialize gBody
for i = 1:length(sensorData.Orientation)
    gBody(i,:) =
    gNED*eul2rotm(deg2rad(sensorData.Orientation(i,:))); %
    applies rotation matrix at each timestep
end

% convert acceleration data from m/s^2 to G:
sensorData.Acceleration = sensorData.Acceleration./9.81;

% remove gravity
sensorData.Acceleration = sensorData.Acceleration - gBody;
```

Root Mean Square. The root mean square of the magnitude of the resulting acceleration graph will be used as the variable for analysis.

$$RMS = \sqrt{\frac{1}{n} \sum_i x_i^2}$$

References

- [1] C. Thai, “Measuring Gait of Knee Arthroplasty Patients Pre-Op and Post-Op Using a Smartphone,” MS Thesis, California Polytechnic State University, San Luis Obispo, 2024.

APPENDIX D: PARTICIPANT DATA, STATIC CONTROL GROUP

Table D-1. Static data of the control group participants.

Participant Code	Sex	Age (years)	Double Leg RMS (m/s ²)	Tandem RMS (m/s ²)	Single Leg RMS (m/s ²)
C1	M	22	0.091952784	0.217843626	0.199074650
C2	F	20	0.100837703	0.103763373	0.117104967
C3	M	22	0.101849039	0.093829502	0.105899126
C4	M	22	0.159314546	0.158412847	0.287778697
C5	F	22	0.074893833	0.079824495	0.088443965
C6	F	22	0.057361174	0.083670909	0.095332002
C7	M	22	0.075135322	0.074787297	0.110937172
C8	M	21	0.074787297	0.123762543	0.234810940
C9	F	21	0.123287198	0.399683798	0.306817083
C10	M	22	0.537691208	0.065606153	0.116899482
C11	M	21	0.261249747	0.409264806	2.995539345
C12	F	21	0.114934748	0.130652165	0.119203221
C13	F	21	0.122556393	0.119445881	0.123827559
C14	M	21	0.224409162	0.196245536	0.115397486
C15	M	21	0.081696274	0.126627908	0.196525882
C16	M	19	0.059985168	0.169016638	0.131171139
C17	M	18	0.271918153	0.239157127	0.185711860
C18	F	21	0.061495175	0.122792086	0.118339363
C19	F	22	0.077599004	0.098625676	0.152527592
C20	F	22	0.029264890	0.055659485	0.053798629

APPENDIX E: PARTICIPANT DATA, STATIC DISC GOLF GROUP

Table E-1. Static data of the disc golf group participants.

Participant Code	Sex	Age (years)	Double Leg RMS (m/s ²)	Tandem RMS (m/s ²)	Single Leg RMS (m/s ²)
D1	M	66	0.659949815	0.190738894	0.143701142
D2	M	41	0.133308990	0.179537205	0.243435592
D3	M	48	0.089821255	0.175331323	2.447772703
D4	M	38	0.092876987	0.186158688	2.562048249
D5	M	21	0.072359202	0.116196651	0.162158570
D6	M	35	0.051622462	0.078741535	0.186852264
D7	M	66	0.146241023	0.285150966	1.799324357
D8	M	27	0.080923140	0.127100655	0.198862822
D9	M	27	0.082244365	0.170827657	0.270640264
D10	M	27	0.184189495	0.178899424	0.799608559
D11	M	28	0.175580629	0.147104487	0.173239985
D12	M	33	0.153646996	0.195892095	0.141294198
D13	M	43	0.137912033	0.220545395	0.282938646
D14	M	51	0.156057476	0.258742294	1.041320877
D15	M	32	0.106168986	0.215731161	1.827514232
D16	M	26	0.084994973	0.186580281	0.394076491
D17	M	25	0.256835398	0.319014690	0.310846903
D18	M	36	0.052811534	0.080890417	0.099897462
D19	M	47	1.146796028	0.367116910	0.237587014
D20	M	30	0.235573146	0.188581733	0.235691311

APPENDIX F: PARTICIPANT DATA, DYNAMIC CONTROL GROUP

Table F-1. Dynamic data of the control group participants.

Participant Code	Mean Gait Speed (ft/s)	Mean Step Width (in)	Mean Stride Length (in)	Mean Cadence (steps/min)	Mean Right Step Length (in)	Mean Left Step Length (in)
C1	3.80	6.00	54.00	101.00	27.00	27.00
C2	4.08	6.67	54.33	108.33	27.67	26.00
C3	3.99	5.67	57.67	99.33	28.67	29.00
C4	3.05	6.00	47.67	92.00	23.33	24.33
C5	3.37	3.33	48.33	100.33	24.33	24.00
C6	4.26	5.67	56.00	109.67	29.00	27.00
C7	5.04	4.33	62.67	116.00	31.67	31.00
C8	3.47	4.33	53.67	93.00	27.00	27.00
C9	3.85	6.67	50.00	110.67	25.00	25.00
C10	4.45	4.00	65.00	98.67	31.67	33.00
C11	4.06	4.00	59.00	99.67	29.33	29.33
C12	4.38	4.00	57.33	110.00	28.67	28.67
C13	4.21	5.67	55.00	110.67	25.33	29.33
C14	4.11	6.00	55.67	106.67	27.67	27.33
C15	4.16	4.00	59.00	101.33	29.67	29.67
C16	4.31	4.33	58.67	105.67	30.67	28.00
C17	4.53	4.67	60.67	107.33	29.00	32.00
C18	4.43	5.67	55.67	114.67	28.00	27.67
C19	3.60	5.67	48.00	107.33	23.67	24.33
C20	4.02	5.00	52.67	110.33	26.00	26.67

APPENDIX G: PARTICIPANT DATA, DYNAMIC DISC GOLF GROUP

Table G-1. Dynamic data of the disc golf group participants.

Participant Code	Mean Gait Speed (ft/s)	Mean Step Width (in)	Mean Stride Length (in)	Mean Cadence (steps/min)	Mean Right Step Length (in)	Mean Left Step Length (in)
D1	4.41	5.33	56.67	111.33	29.67	27.33
D2	4.62	5.67	64.33	103.67	32.00	32.33
D3	4.22	5.33	57.33	106.00	29.00	28.00
D4	4.03	6.33	57.67	100.33	26.33	31.00
D5	4.52	5.00	59.00	110.00	30.33	28.67
D6	3.70	6.00	52.00	102.33	26.67	25.67
D7	4.32	6.33	52.67	118.00	27.33	25.67
D8	4.56	4.00	63.67	103.67	31.67	31.33
D9	4.33	6.67	62.67	99.33	30.67	32.00
D10	4.41	5.67	57.33	110.67	28.33	29.00
D11	4.67	4.00	60.00	112.00	29.67	30.67
D12	4.86	6.00	62.67	111.67	32.33	30.33
D13	4.29	5.33	59.00	104.33	29.33	30.00
D14	4.43	7.00	60.33	106.67	30.00	30.33
D15	4.23	5.67	58.33	104.67	30.00	28.33
D16	4.04	5.67	56.67	102.67	26.33	30.33
D17	4.73	5.00	63.00	108.00	32.00	31.00
D18	4.61	3.67	67.33	99.00	34.33	32.67
D19	4.33	6.67	58.00	107.67	30.00	27.67
D20	4.55	6.00	59.33	110.33	30.00	29.33

APPENDIX H: BAR GRAPHS OF RESULTS

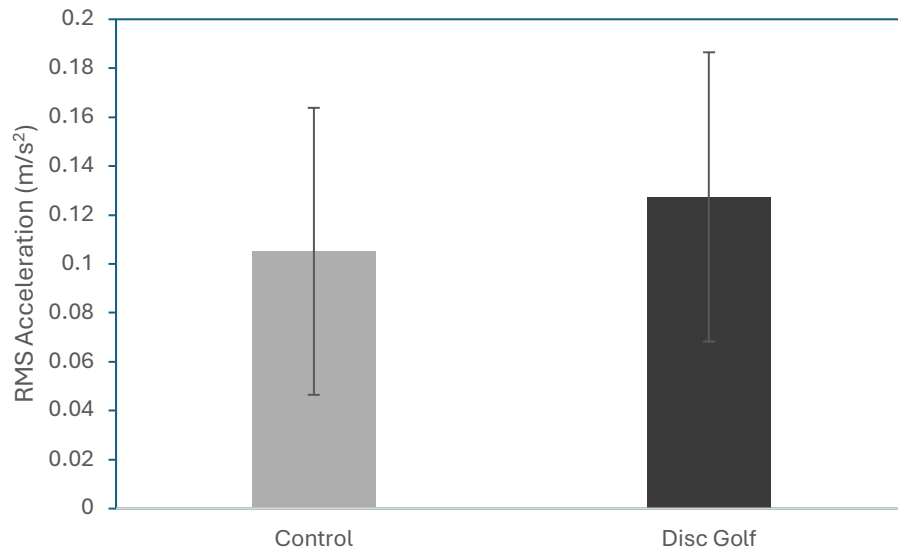


Figure H-1. RMS acceleration in the double leg stance of both groups ($p=0.8676$)

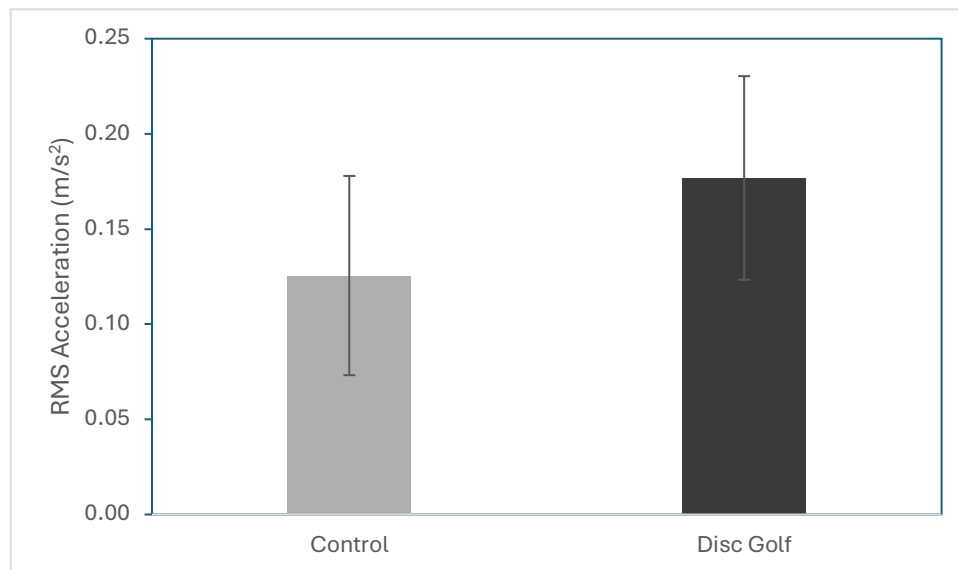


Figure H-2. RMS acceleration in the tandem stance of both groups ($p=0.9968$)

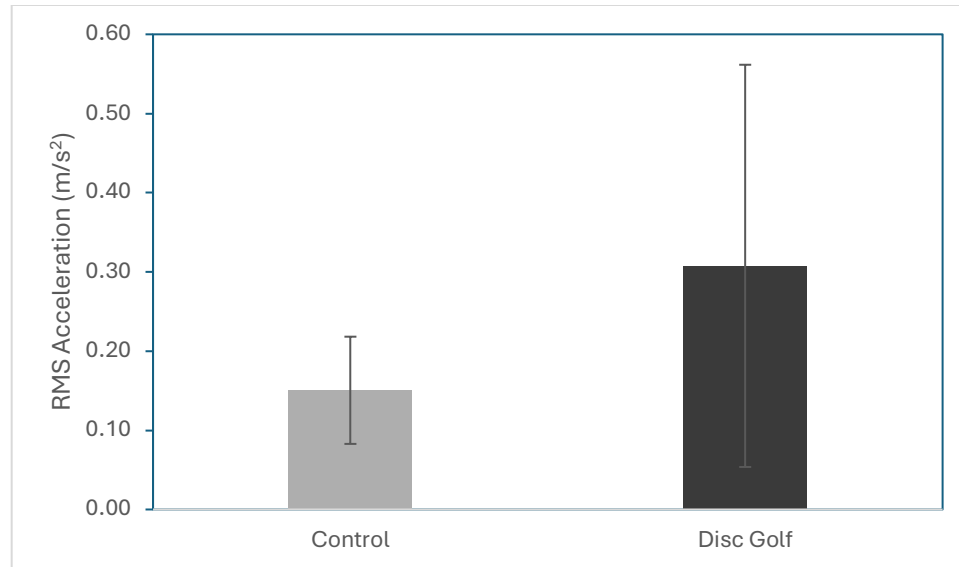


Figure H-3. RMS acceleration in the single leg stance of both groups ($p=0.9891$).

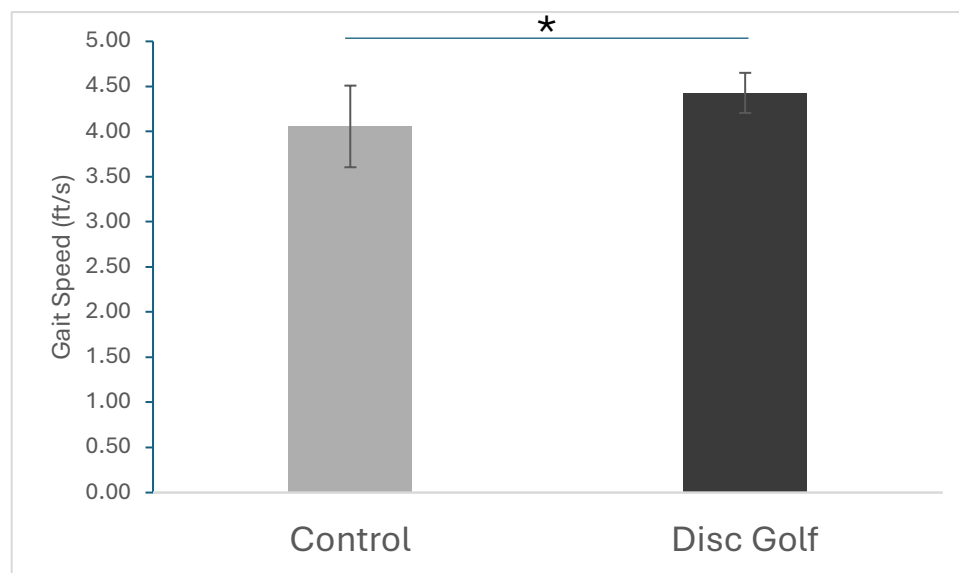


Figure H-4. Mean gait speed of both groups ($p=0.0014$) * = significant difference ($p<0.05$)

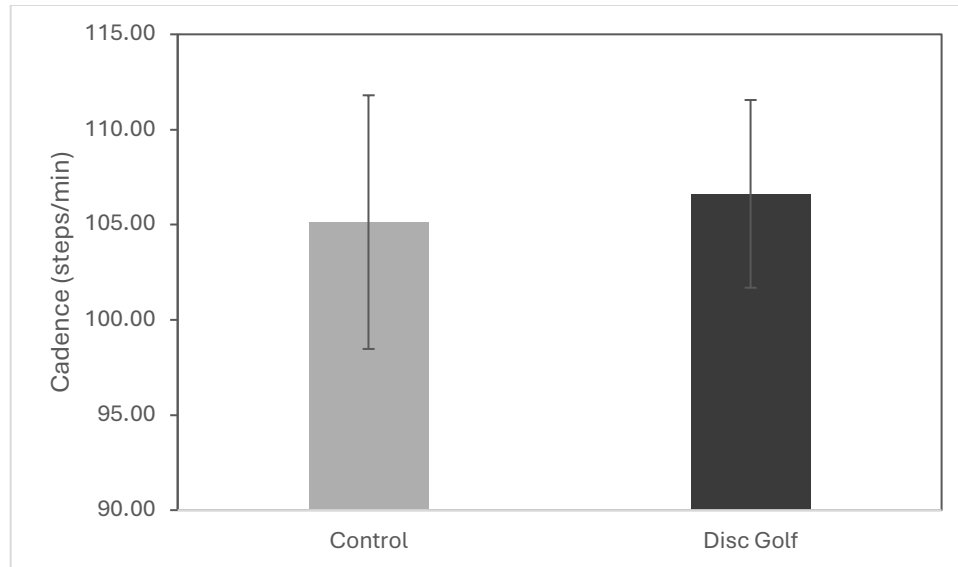


Figure H-5. Mean cadence of both groups ($p=0.2145$).

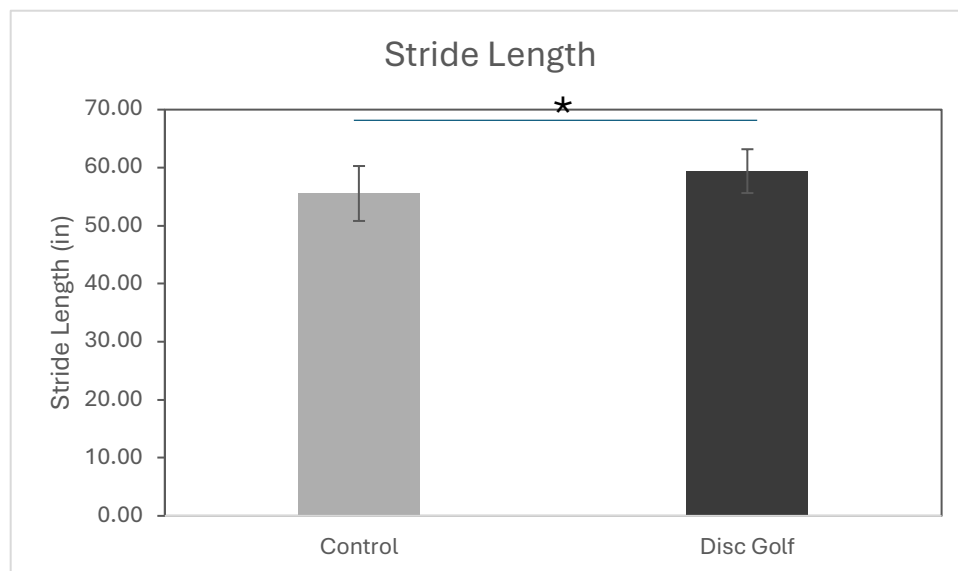


Figure H-6. Mean stride length of both groups ($p=0.0037$) * = significant difference ($p<0.05$)

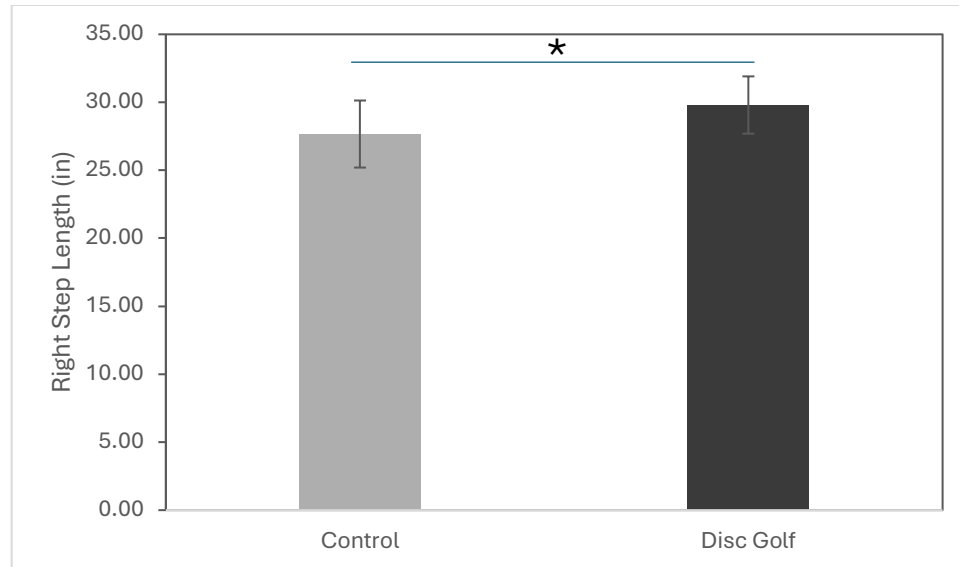


Figure H-7. Mean right step length of both groups ($p=0.0028$). * = significant difference ($p<0.05$)

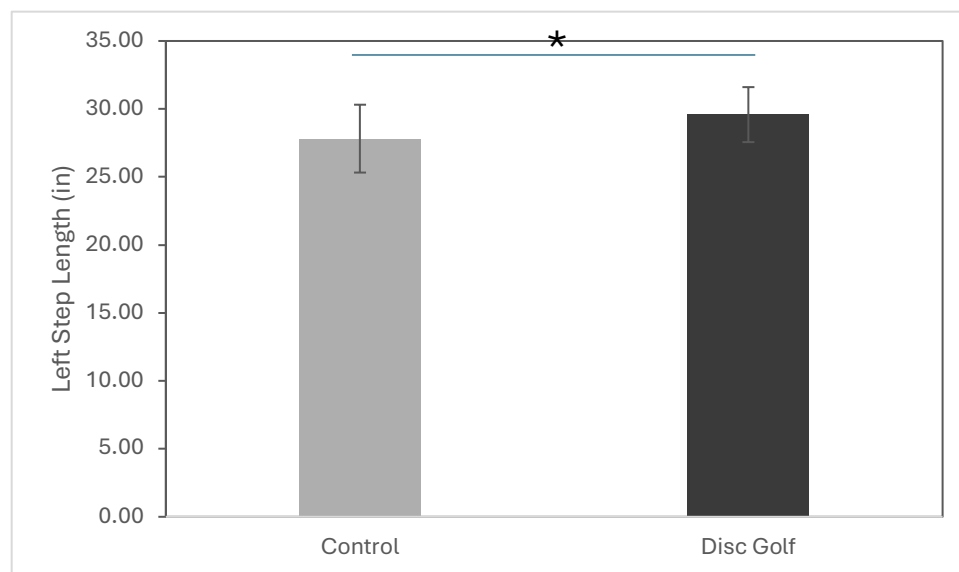


Figure H-8. Mean left step length of both groups ($p=0.0094$). * = significant difference ($p<0.05$)

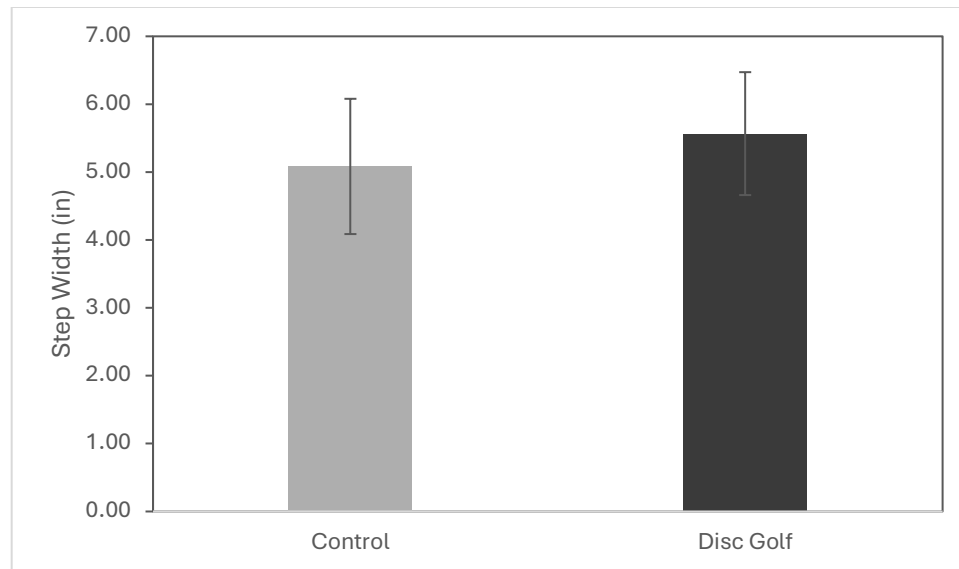


Figure H-9. Mean step width of both groups ($p=0.9413$).